Mid-term Project — Track B (The Strategist)

Rongwen Weng · GitHub: rongwenweng-arch · Strategy: **cross-family-router**

Wallet

`0x7Bfe6bef0E8092bf2d829a75Af11dDfd6A7b4140`

Network: Base Sepolia. Files: **strategy.py** (the strategy, **decide(view)**), my_strategy_bot.py (on-chain adapter), NOTES.md. Pull Request: https://github.com/aaron-recompile/AI_Finance/pull/14

What my strategy does and why it should make money

The arena quotes MAX/tUSDC on four venues that belong to two families: the flash-demo AMMs (CHEAP, DEAR) and the Uniswap-v2 style pools (PoolA, PoolB). The provided bots only arbitrage inside one family, so the price gap *between* the families was left open (about 15.6% when I started: PoolA ≈ 1750 vs DEAR ≈ 2023 tUSDC per MAX, against only 0.6% of total swap fees). My bot treats all four venues as one market: it tries every ordered (buy venue, sell venue) pair, buys MAX where it is cheapest, then re-reads the chain and sells on whichever venue pays the most at that moment. Trade size is not a grid search; chaining two constant-product swaps gives out(D) = aD / (b + cD), so profit peaks at D* = (√(ab) − b) / c, and I trade min(D*, my balance). It can lose because the buy and the sell are two separate transactions: if the price moves between them I may sell for less than I paid, and my own trades close the gap, so profit per round shrinks (the gap fell from 15.6% to 5.6% over seven rounds).

Transactions (Basescan)

#	Route	Profit (tUSDC)	Sell transaction
1	PoolA → DEAR	+307.08	`0x7da3974d5f1cbb063aa37231fa9ea2219099ad6221b55cd0d7a58cb7c05045e7`
2	PoolB → DEAR	+320.88	`0xb319a4885550913443fee41d7a7f211760dd11f891bedf34c884a2736dd06e32`
3	PoolA → DEAR	+333.86	`0x23d75d1c26e109d159921a5f2c6293821f75b745650c005e8bbebeb708fd4c72`
4	PoolB → DEAR	+309.55	`0x9ac8b21f8976d9ba7599f8c1fead45850a96c37a5d90f54c49a3670e6c5026f3`
5	PoolA → DEAR	+318.88	`0xc85ca3b9be89b1a46c72b0922f6b240e59a631f6154208874d887f41500f5b28`
6	PoolA → DEAR	+242.90	`0x58d4ac405aedd21c998f0e6ca7640fbb5311b70ea6ab706a9b342deb827a7018`
7	PoolA → DEAR	+241.91	`0x89f0f3d496858237c9831e3520c79d8133f581f52b9aa1af73c2e34e8774fc93`

Round 1 was a single direct run (buy tx `0x466585f303eac3dc6bb506f6fa7c58d582cd9c15e91a33d5b28d4448ce116cd1`). Rounds 2–6 ran under the course orchestrator and are the five fires on the leaderboard below (+1,526.07 tUSDC). Round 7 was interrupted between its buy and its sell when I stopped the orchestrator, which left 2.1312 MAX unsold; I added an **--unwind** option to the bot and used it to sell that leg on DEAR. Wallet over all seven rounds: 2,193.46 → 4,268.52 tUSDC (**+2,075.06**).

Leaderboard

